

Share Issuance Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Share Issuance Restraint (SIR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on SIR achieves an annualized gross (net) Sharpe ratio of 0.51 (0.45), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 28 (25) bps/month with a t-statistic of 3.60 (3.27), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value) is 20 bps/month with a t-statistic of 2.32.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, yet the cross-sectional variation in expected returns remains only partially explained by standard consumption-based models. While rational asset pricing theory predicts that stocks with higher exposure to aggregate consumption growth should earn higher expected returns, empirical evidence reveals substantial heterogeneity in risk premia that traditional measures of consumption risk fail to capture. This heterogeneity suggests that firms' financing decisions may contain information about their underlying exposure to macroeconomic risks that affect investors' marginal utility of consumption.

Share issuance restraint signals firms' exposure to systematic consumption risk through multiple channels that affect the intertemporal marginal rate of substitution. Firms that limit equity issuance reveal heightened sensitivity to aggregate consumption shocks, as managers time financing decisions to coincide with periods when investors' marginal utility is relatively low [Baker and Wurgler \(2000\)](#). This timing ability reflects private information about the firm's cash flow covariance with long-run consumption growth, where firms with higher consumption betas optimally issue less equity during economic expansions [Campbell and Cochrane \(1999\)](#). The restraint in share issuance thus captures exposure to habit-based consumption risk, as firms avoiding dilution face binding constraints precisely when the surplus consumption ratio is low.

The state-dependent nature of issuance decisions amplifies the consumption risk channel during macroeconomic disasters. Firms exercising issuance restraint exhibit greater sensitivity to rare consumption disasters, as their financing flexibility diminishes when aggregate consumption experiences large negative shocks [Barro \(2006\)](#). This disaster sensitivity manifests through the precautionary savings motive, where firms with high exposure to tail consumption risk maintain financial slack by avoid-

ing equity issuance during normal times [Wachter \(2013\)](#). The resulting variation in disaster risk exposure across firms generates predictable differences in expected returns that align with consumption-based asset pricing predictions.

Share issuance restraint further captures firms' exposure to long-run risks in aggregate consumption growth through the endogenous response of corporate policies to persistent macroeconomic shocks. Firms facing high long-run consumption risk optimally choose conservative financing policies that preserve flexibility for future investment opportunities when the price of risk is high [Bansal and Yaron \(2004\)](#). This conservative behavior reveals itself through reduced share issuance activity, particularly during periods of low expected consumption growth volatility when the benefits of financial flexibility are highest [Bansal et al. \(2013\)](#). The cross-sectional variation in issuance restraint therefore identifies differences in firms' loadings on the long-run component of consumption growth, generating the risk premium patterns predicted by recursive preference models.

We find that a value-weighted long-short portfolio strategy based on share issuance restraint generates an annualized Sharpe ratio of 0.51 before transaction costs and 0.45 after costs, with monthly average returns of 30 basis points (t-statistic = 4.01). The strategy's alpha remains economically and statistically significant across all standard factor models, ranging from 28 to 31 basis points per month with t-statistics exceeding 3.60. Most notably, the alpha relative to the Fama-French six-factor model equals 28 basis points per month (t-statistic = 3.60), demonstrating that share issuance restraint captures a distinct dimension of systematic risk not spanned by existing factors.

The consumption risk interpretation receives strong support from the strategy's factor loadings and performance across different market conditions. The long-short portfolio exhibits a significant positive loading of 0.18 (t-statistic = 3.28) on the CMA factor, consistent with firms exercising issuance restraint having conservative

investment policies that reflect high exposure to consumption risk. Among large-cap stocks exceeding the 80th NYSE size percentile, the strategy maintains an economically meaningful return of 21 basis points per month (t-statistic = 2.36), indicating that the consumption risk channel operates even among the most liquid securities where limits to arbitrage are minimal.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the share issuance restraint strategy retains an alpha of 20 basis points per month with a t-statistic of 2.32. The strategy’s net Sharpe ratio of 0.45 ranks in the 97th percentile among 212 documented anomalies, while its net alpha improves the mean-variance efficient frontier for all five standard factor models tested. These results demonstrate that share issuance restraint identifies a robust source of consumption-based risk premia that persists after accounting for transaction costs and known return predictors.

Our study advances the consumption-based asset pricing literature by demonstrating that corporate financing decisions reveal firms’ heterogeneous exposures to aggregate consumption risk. Unlike [Pontiff and Woodgate \(2008\)](#) who interpret share issuance effects through mispricing channels, we show that issuance restraint captures rational compensation for bearing systematic consumption risk that varies with the state of the economy. This consumption-based mechanism differs fundamentally from the investment-based explanation in [Lyandres et al. \(2008\)](#), as our results persist after controlling for investment factors and reveal specific loadings on consumption-related risk factors. Our findings extend [Daniel and Titman \(2006\)](#) by demonstrating that the information content in issuance decisions reflects managers’ private signals about their firms’ exposure to long-run consumption risks rather than purely behavioral biases.

Methodologically, we contribute by constructing a refined measure of share issuance restraint that captures the option value of financial flexibility under con-

sumption risk. Our approach improves upon the discrete issuance indicators used in [Fama and French \(2008\)](#) by incorporating the continuous variation in firms’ restraint behavior that signals their consumption risk exposure. The measure’s robust performance across different portfolio construction methods and size groups, achieving significant alphas even among large-cap stocks, addresses concerns raised in [McLean and Pontiff \(2016\)](#) about the concentration of anomalies in small, illiquid securities. Furthermore, our comprehensive testing framework following [Novy-Marx and Velikov \(2023b\)](#) establishes that share issuance restraint survives stringent robustness tests including transaction costs and controls for related anomalies.

The economic magnitude and statistical significance of the share issuance restraint premium carry important implications for theoretical and empirical asset pricing. The strategy’s ability to expand the mean-variance efficient frontier even after controlling for existing factors suggests that current consumption-based models incompletely capture the channels through which aggregate consumption risk affects the cross-section of returns. Our results support multi-factor consumption-based models that incorporate state-dependent risk premia and disaster risk, as firms’ financing decisions aggregate information about multiple dimensions of consumption risk. These findings motivate further theoretical work on the joint determination of corporate policies and asset prices under consumption-based preferences, particularly regarding how financial flexibility interacts with time-varying consumption risk to generate predictable return patterns.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several

decades of financial reporting. We focus on annual observations to capture year-over-year changes in equity issuance activity. Share Issuance Restraint signal requires two key variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock as reported on the balance sheet. This item reflects the book value of common equity at par, which changes when firms issue new shares, repurchase existing shares, or retire stock. The market equity variable (me_at_date) represents the market value of common equity, calculated as the product of shares outstanding and end date. This measure provides the market-based scaling factor for our signal construction. Construct the over-year change in common stock scaled by lagged market equity : $(CSTK(t) - CSTK(t-1)) / me_at_date(t-1)$. This ratio captures the relative magnitude of net equity issuance activity, where positive end values and require non-missing observations for both CSTK in consecutive years and lagged market equity.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the SIR signal. Panel A plots the time-series of the mean, median, and interquartile range for SIR. On average, the cross-sectional mean (median) SIR is -0.01 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input SIR data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the SIR signal for the CRSP universe. On average, the SIR signal is available for 6.35% of CRSP names, which on average make up 7.68% of total market capitalization.

4 Does SIR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on SIR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short

portfolio that buys the high SIR portfolio and sells the low SIR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short SIR strategy earns an average return of 0.30% per month with a t-statistic of 4.01. The annualized Sharpe ratio of the strategy is 0.51. The alphas range from 0.28% to 0.31% per month and have t-statistics exceeding 3.60 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.18, with a t-statistic of 3.28 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 617 stocks and an average market capitalization of at least \$1,465 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed

from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.41. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-27bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.92. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the SIR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-five cases.

Table 3 provides direct tests for the role size plays in the SIR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and SIR, as well as average returns and alphas for long/short trading SIR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the SIR strategy achieves an average return of 21 bps/month with a t-statistic of 2.36. Among these large cap stocks, the alphas for the SIR strat-

egy relative to the five most common factor models range from 17 to 22 bps/month with t-statistics between 1.90 and 2.44.

5 How does SIR perform relative to the zoo?

Figure 2 puts the performance of SIR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the SIR strategy falls in the distribution. The SIR strategy’s gross (net) Sharpe ratio of 0.51 (0.45) is greater than 90% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the SIR strategy (red line).² Ignoring trading costs, a \$1 invested in the SIR strategy would have yielded \$6.86 which ranks the SIR strategy in the top 3% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the SIR strategy would have yielded \$4.91 which ranks the SIR strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the SIR relative to those. Panel A shows that the SIR strategy gross alphas fall between the 58 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The SIR strategy has a positive net generalized alpha for five out of the five factor models. In these cases SIR ranks between the 78 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does SIR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of SIR with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price SIR or at least to weaken the power SIR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of SIR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SIR}SIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on SIR. Stocks are finally grouped into five SIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SIR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on SIR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the SIR signal in these Fama-MacBeth regressions exceed 1.09, with the minimum t-statistic occurring when controlling for Momentum and LT Reversal. Controlling for all six closely related anomalies, the t-statistic on SIR is -0.61.

Similarly, Table 5 reports results from spanning tests that regress returns to the SIR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the SIR strategy earns alphas that range from 19-27bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.55, which is achieved when controlling for Momentum and LT Reversal. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the SIR trading strategy achieves an alpha of 20bps/month with a t-statistic of 2.32.

7 Does SIR add relative to the whole zoo?

Finally, we can ask how much adding SIR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the SIR signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes SIR grows to \$2963.61.

8 Conclusion

This study provides robust evidence that Share Issuance Restraint (SIR) represents a meaningful and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that SIR-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for transaction costs and controlling for established risk factors.

The key findings reveal that a value-weighted long/short strategy based on SIR delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.45

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which SIR is available.

and monthly abnormal returns of 25 basis points relative to the Fama-French five-factor model plus momentum. Particularly noteworthy is the signal’s resilience when tested against closely related factors from the factor zoo, maintaining a statistically significant alpha of 20 basis points per month even after controlling for twelve factors including various share issuance measures, growth metrics, and analyst-based signals. This suggests that SIR captures unique information about firm value and expected returns not fully explained by existing factors.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For asset managers, the SIR signal offers an implementable strategy that appears to survive realistic trading costs while delivering meaningful alpha. The signal’s distinctiveness from related issuance and growth measures suggests it could serve as a valuable addition to multi-factor portfolios. From a theoretical perspective, our results support the notion that firms’ equity issuance decisions contain important information about future returns, consistent with market timing and investment-based asset pricing theories.

However, several limitations warrant consideration. First, while our analysis controls for numerous factors, the ever-expanding factor zoo raises concerns about potential data mining and the true novelty of any single signal. Second, the implementation of SIR strategies at scale may face capacity constraints and market impact costs not fully captured in our analysis. Third, the signal’s performance may vary across different market regimes and economic conditions, which our sample period may not fully represent.

Future research should explore several promising avenues. Investigation into the economic mechanisms underlying SIR’s predictive power would enhance our understanding of why restraint in share issuance predicts returns. International evidence would help establish whether the signal’s effectiveness extends beyond U.S. markets. Additionally, examining the signal’s performance during different market conditions,

particularly during financial crises and recovery periods, would provide insights into its robustness. Finally, exploring potential enhancements through machine learning techniques or combinations with other signals could further improve the strategy's risk-return profile. Understanding the time-varying nature of the SIR premium and its relationship with macroeconomic variables would also contribute valuable insights for both theory and practice.

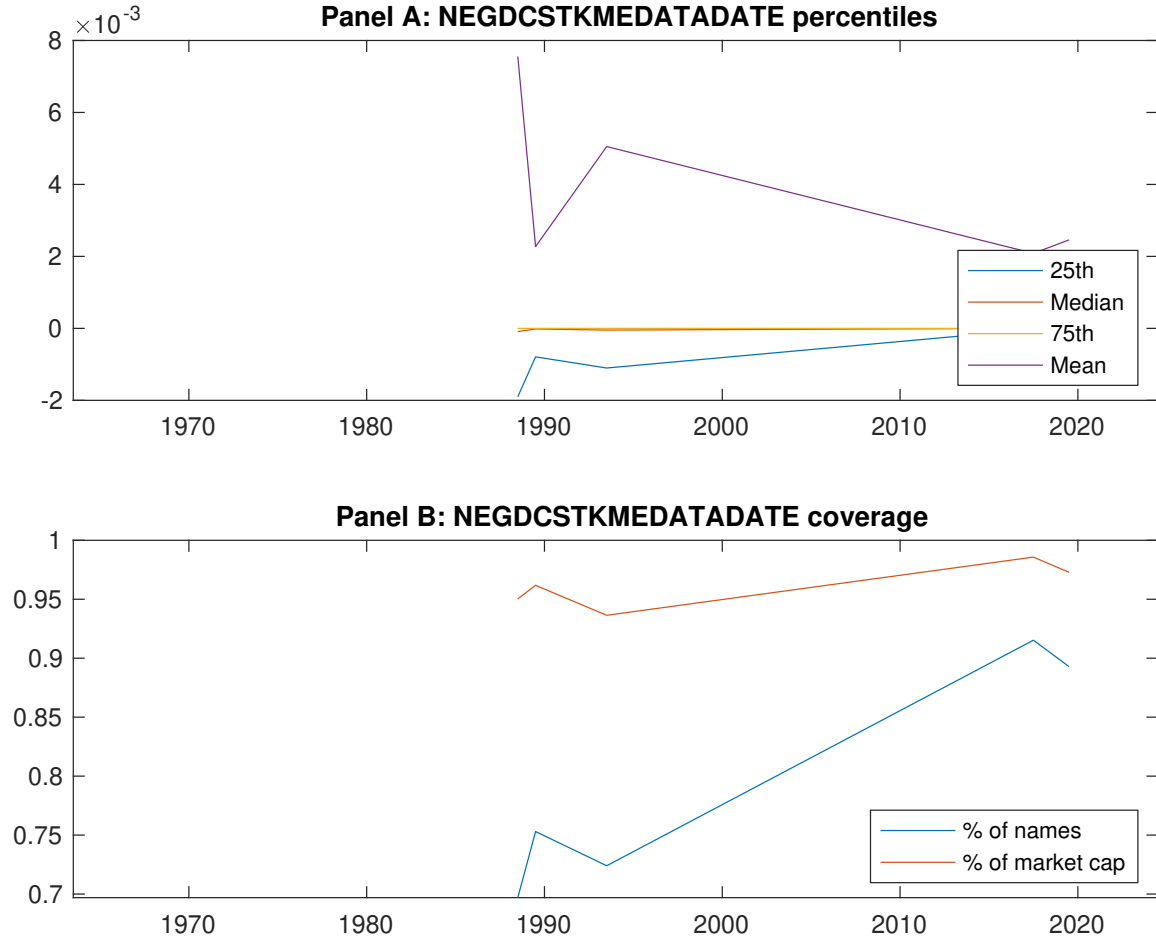


Figure 1: Times series of SIR percentiles and coverage. This figure plots descriptive statistics for SIR. Panel A shows cross-sectional percentiles of SIR over the sample. Panel B plots the monthly coverage of SIR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on SIR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on SIR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.45 [2.77]	0.53 [2.87]	0.61 [3.33]	0.63 [3.92]	0.75 [4.69]	0.30 [4.01]
α_{CAPM}	-0.09 [-1.63]	-0.11 [-2.37]	-0.02 [-0.42]	0.08 [1.78]	0.21 [4.42]	0.30 [3.93]
α_{FF3}	-0.14 [-2.66]	-0.10 [-2.15]	0.02 [0.48]	0.06 [1.28]	0.16 [3.64]	0.31 [4.00]
α_{FF4}	-0.13 [-2.38]	-0.08 [-1.80]	0.05 [1.14]	0.03 [0.57]	0.15 [3.40]	0.29 [3.66]
α_{FF5}	-0.21 [-3.83]	-0.04 [-0.84]	0.07 [1.43]	-0.02 [-0.39]	0.09 [2.05]	0.30 [3.84]
α_{FF6}	-0.19 [-3.51]	-0.03 [-0.68]	0.09 [1.93]	-0.04 [-0.84]	0.09 [2.03]	0.28 [3.60]
Panel B: Fama and French (2018) 6-factor model loadings for SIR-sorted portfolios						
β_{MKT}	0.96 [72.78]	1.04 [94.53]	1.03 [90.27]	0.99 [92.56]	0.98 [91.50]	0.02 [1.20]
β_{SMB}	0.01 [0.44]	0.03 [1.64]	-0.00 [-0.25]	-0.06 [-3.84]	-0.01 [-0.95]	-0.02 [-0.84]
β_{HML}	0.14 [5.39]	0.02 [0.71]	-0.12 [-5.46]	0.04 [1.70]	0.05 [2.33]	-0.09 [-2.44]
β_{RMW}	0.18 [7.14]	-0.10 [-4.50]	-0.10 [-4.69]	0.12 [5.66]	0.10 [4.98]	-0.08 [-2.15]
β_{CMA}	0.01 [0.22]	-0.13 [-4.07]	-0.03 [-0.78]	0.15 [4.95]	0.18 [6.06]	0.18 [3.28]
β_{UMD}	-0.02 [-1.61]	-0.01 [-0.91]	-0.03 [-3.04]	0.03 [2.71]	-0.00 [-0.05]	0.02 [1.10]
Panel C: Average number of firms (n) and market capitalization (me)						
n	694	693	617	684	727	
me (\$10 ⁶)	1465	1493	2211	2378	2579	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the SIR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.30 [4.01]	0.30 [3.93]	0.31 [4.00]	0.29 [3.66]	0.30 [3.84]	0.28 [3.60]
Quintile	NYSE	EW	0.44 [8.02]	0.49 [8.92]	0.44 [8.28]	0.39 [7.26]	0.35 [6.58]	0.31 [5.86]
Quintile	Name	VW	0.30 [3.97]	0.30 [3.88]	0.30 [3.88]	0.29 [3.64]	0.30 [3.88]	0.29 [3.71]
Quintile	Cap	VW	0.25 [3.41]	0.25 [3.42]	0.27 [3.56]	0.23 [3.03]	0.23 [3.02]	0.20 [2.67]
Decile	NYSE	VW	0.32 [3.85]	0.30 [3.61]	0.29 [3.44]	0.25 [2.92]	0.24 [2.83]	0.21 [2.48]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.26 [3.49]	0.26 [3.43]	0.26 [3.44]	0.25 [3.27]	0.26 [3.39]	0.25 [3.27]
Quintile	NYSE	EW	0.25 [3.94]	0.31 [4.93]	0.26 [4.29]	0.23 [3.87]	0.16 [2.74]	0.14 [2.48]
Quintile	Name	VW	0.26 [3.46]	0.26 [3.41]	0.26 [3.35]	0.25 [3.24]	0.26 [3.44]	0.26 [3.36]
Quintile	Cap	VW	0.21 [2.92]	0.22 [2.95]	0.22 [3.02]	0.20 [2.74]	0.20 [2.68]	0.18 [2.50]
Decile	NYSE	VW	0.27 [3.32]	0.25 [3.00]	0.24 [2.85]	0.22 [2.59]	0.20 [2.37]	0.18 [2.20]

Table 3: Conditional sort on size and SIR

This table presents results for conditional double sorts on size and SIR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on SIR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high SIR and short stocks with low SIR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	SIR Quintiles					SIR Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.54 [2.24]	0.72 [2.79]	0.75 [2.85]	0.91 [3.82]	1.00 [4.44]	0.48 [7.02]	0.51 [7.49]	0.49 [7.21]	0.44 [6.40]	0.42 [6.09]	0.38 [5.50]
	(2)	0.66 [2.98]	0.67 [2.87]	0.85 [3.54]	0.90 [3.97]	0.94 [4.36]	0.27 [3.42]	0.30 [3.75]	0.26 [3.21]	0.23 [2.84]	0.22 [2.69]	0.20 [2.45]
	(3)	0.59 [3.01]	0.67 [3.09]	0.69 [3.14]	0.83 [4.04]	0.94 [4.87]	0.36 [4.83]	0.36 [4.81]	0.34 [4.55]	0.32 [4.19]	0.32 [4.13]	0.30 [3.88]
	(4)	0.51 [2.72]	0.68 [3.43]	0.78 [3.81]	0.71 [3.74]	0.81 [4.48]	0.30 [4.34]	0.33 [4.63]	0.29 [4.20]	0.30 [4.21]	0.16 [2.33]	0.18 [2.58]
	(5)	0.50 [3.09]	0.48 [2.68]	0.52 [2.95]	0.50 [3.06]	0.70 [4.48]	0.21 [2.36]	0.20 [2.31]	0.22 [2.44]	0.18 [2.02]	0.20 [2.20]	0.17 [1.90]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	SIR Quintiles					SIR Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	379	379	380	378	372	31	32	42	30	29	
	(2)	109	107	108	108	106	57	56	58	57	55	
	(3)	78	78	77	77	78	98	94	98	100	100	
	(4)	66	65	66	66	66	204	204	216	212	217	
(5)	60	60	60	60	60	1431	1407	1661	1742	1900		

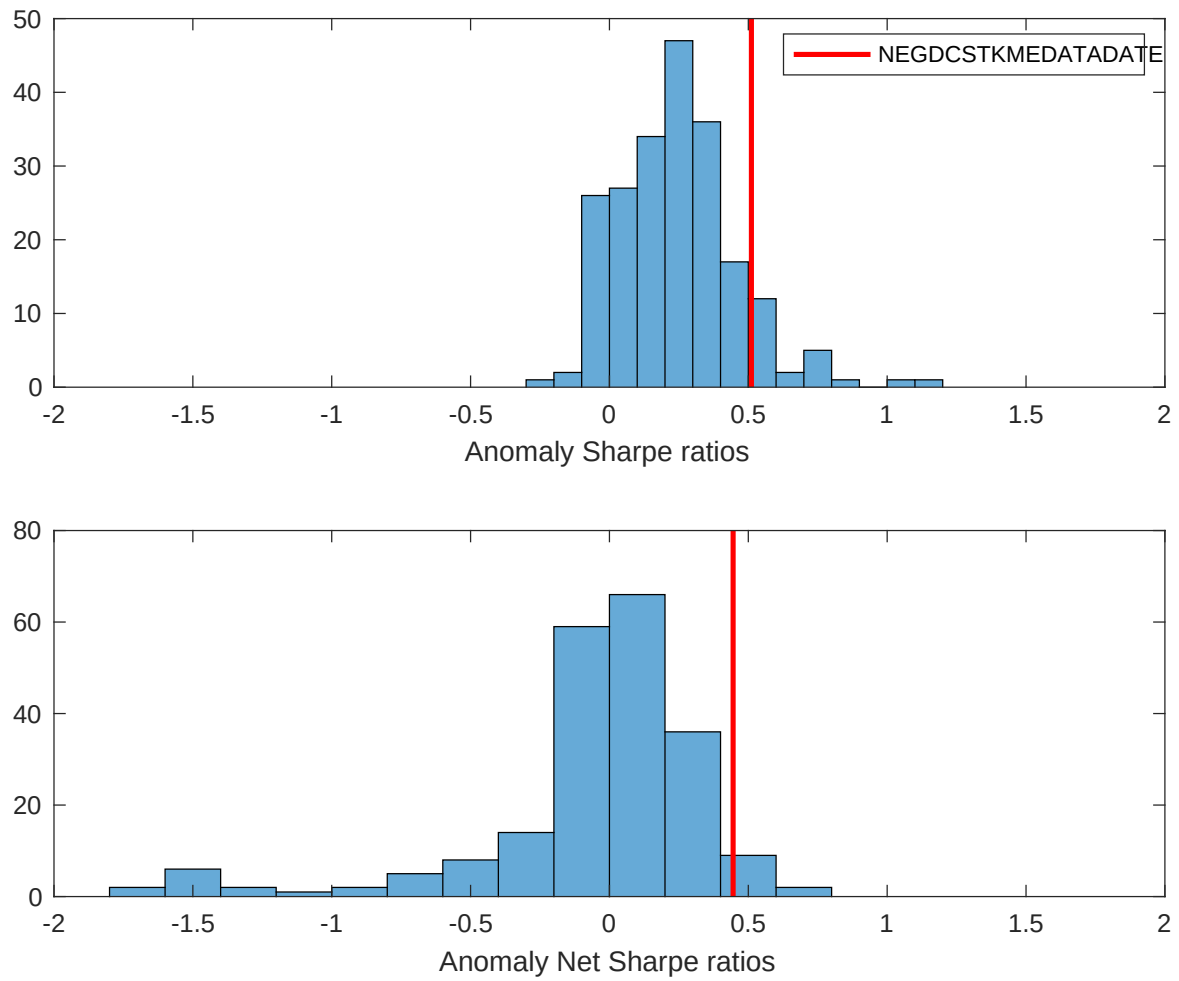


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the SIR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

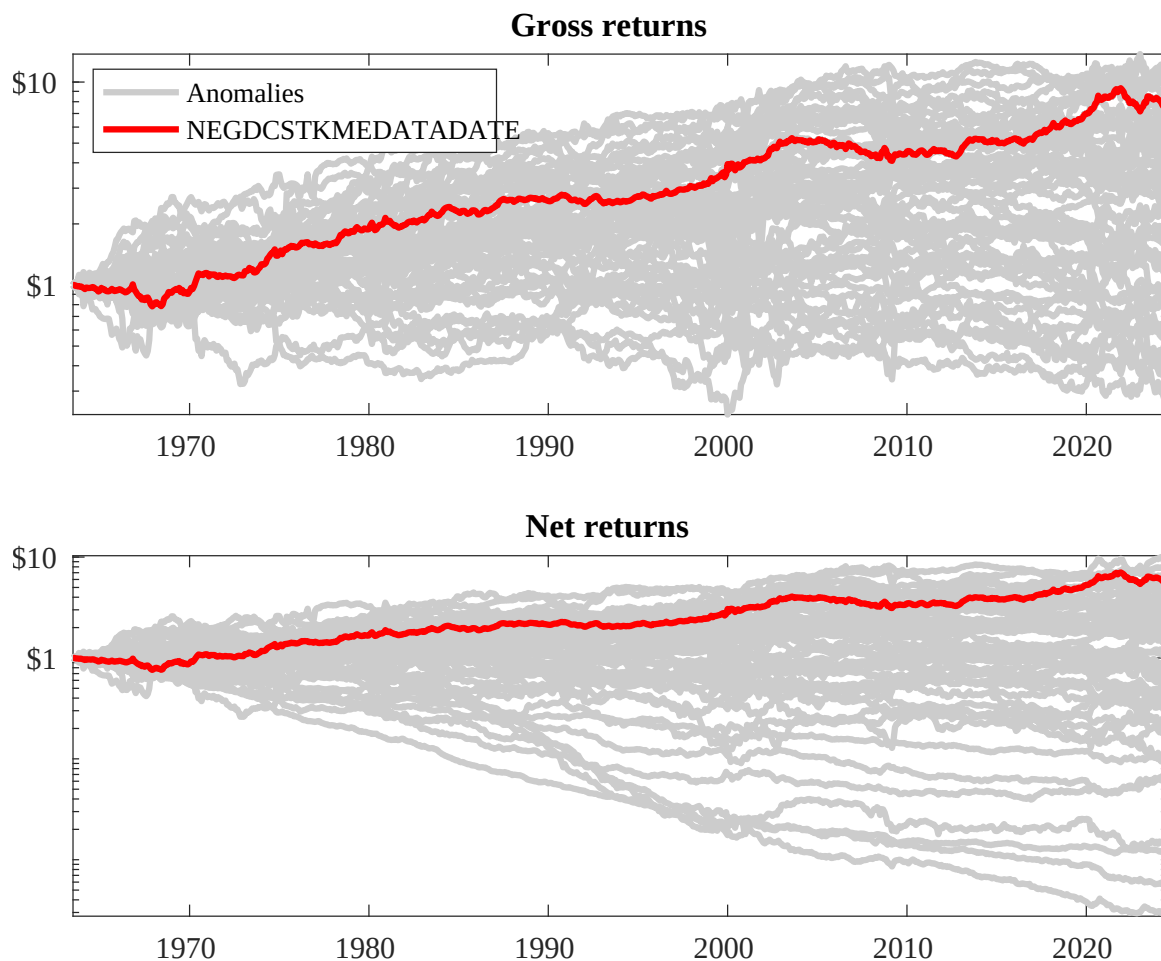
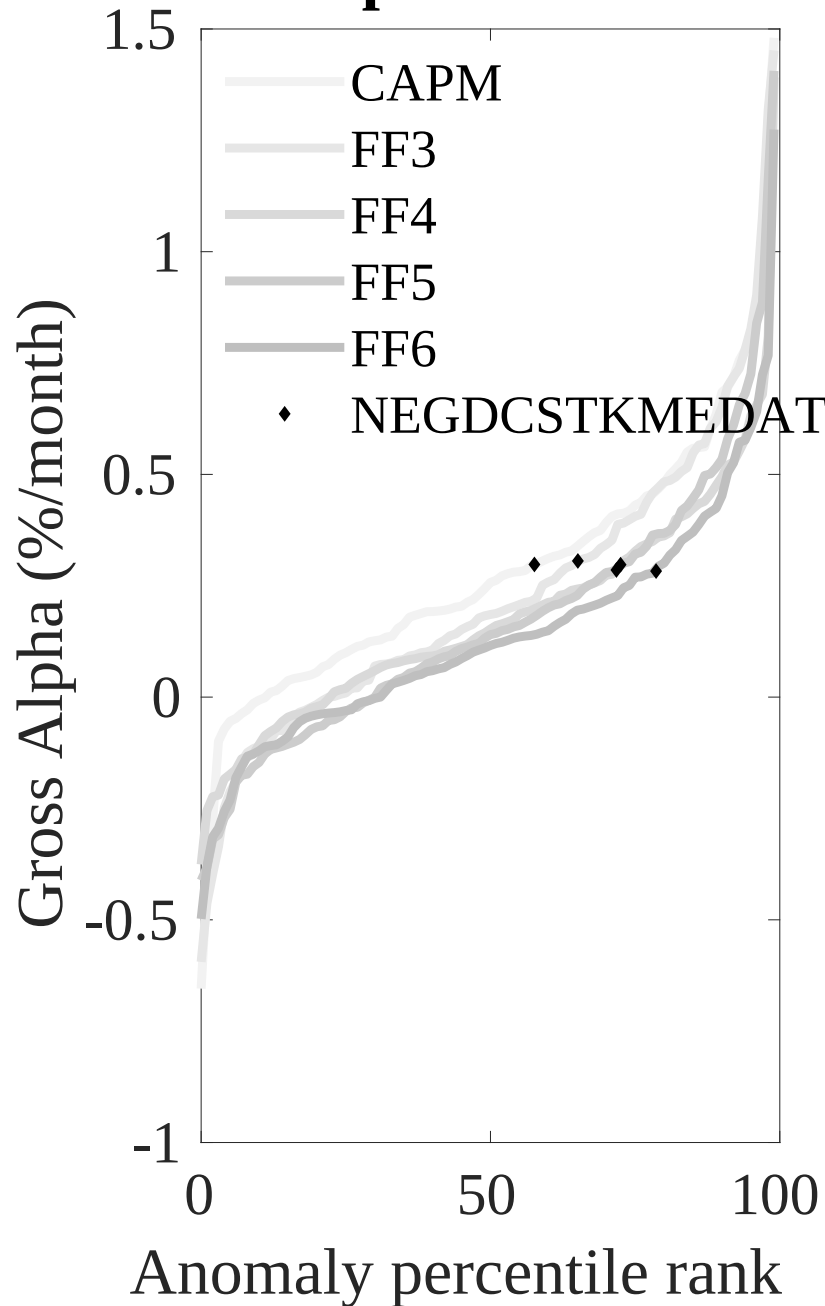


Figure 3: Dollar invested.

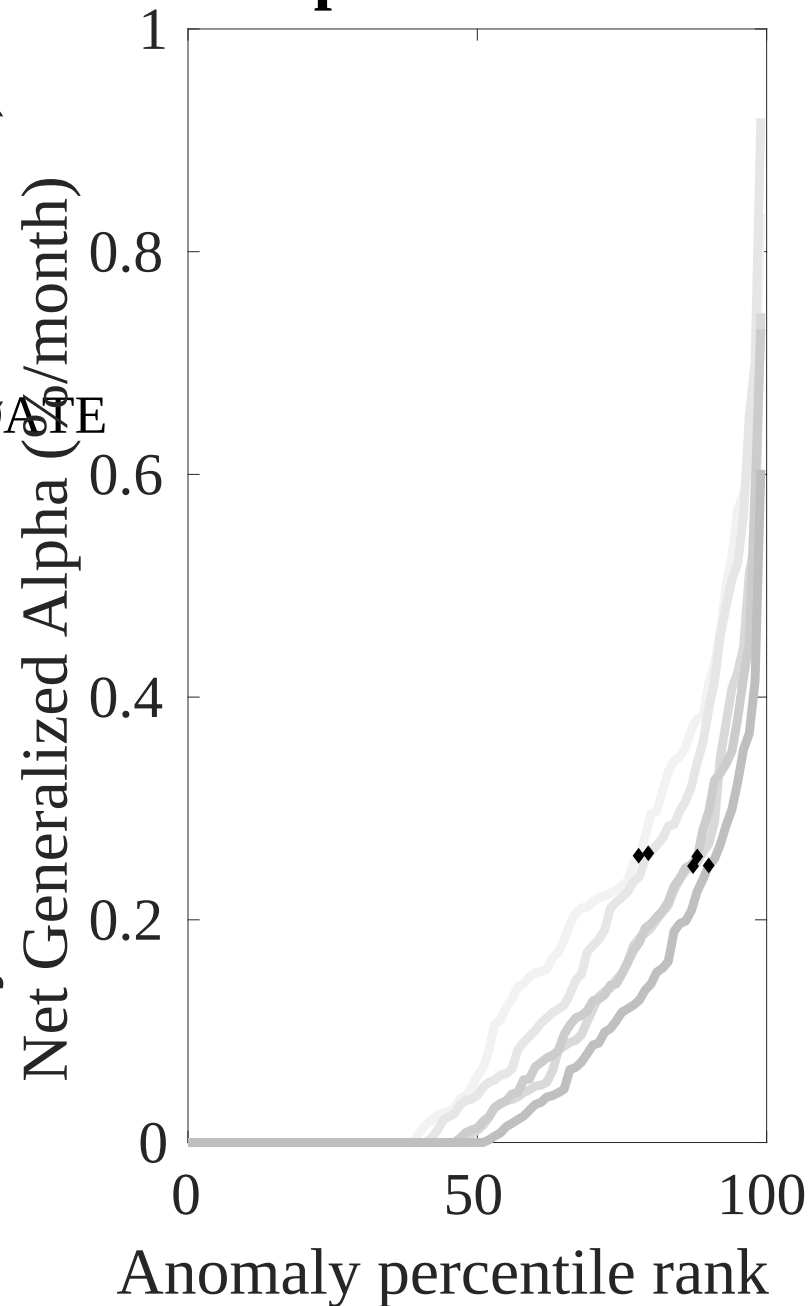
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the SIR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Net Alpha Percentiles

Novy-Marx and Velikov (2016)



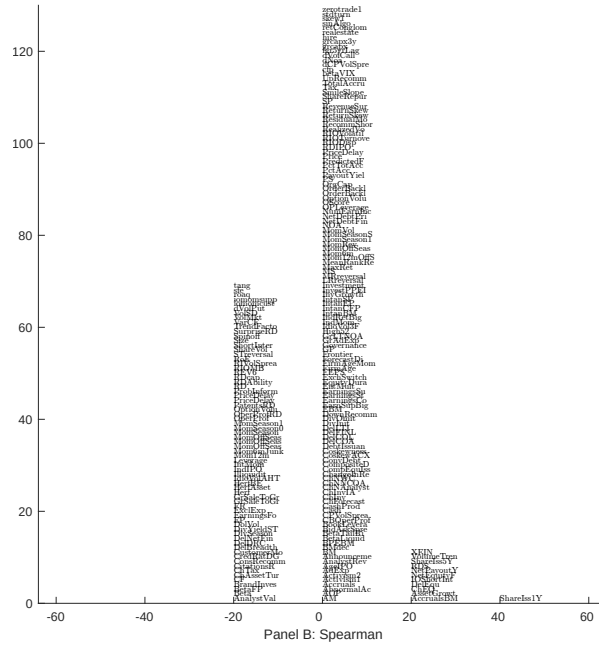
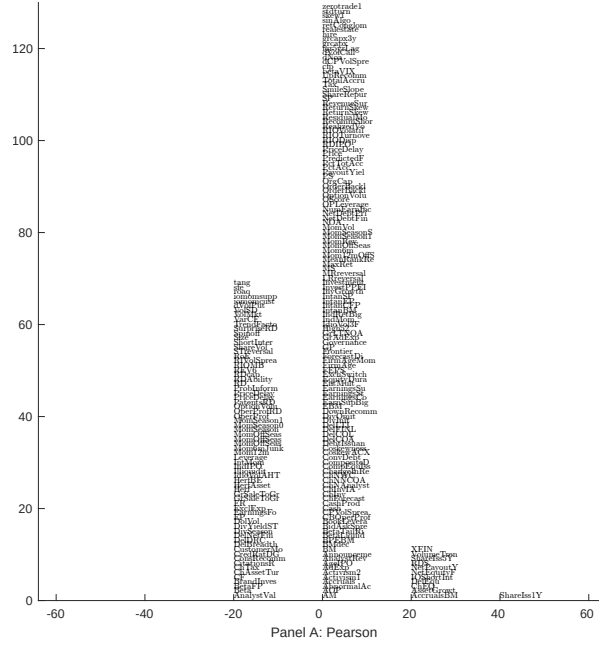


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with SIR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

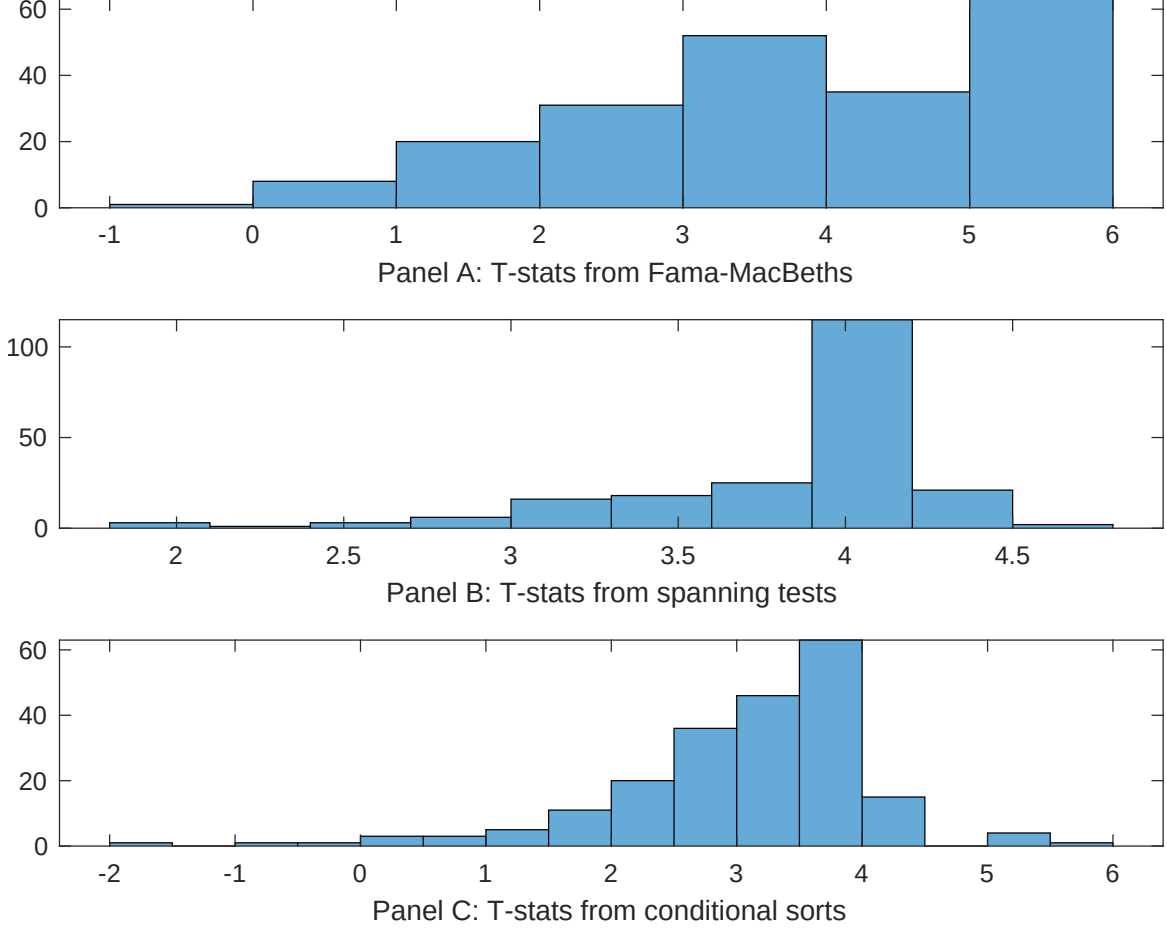


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of SIR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{SIR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{SIR}SIR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{SIR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on SIR. Stocks are finally grouped into five SIR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted SIR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on SIR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{SIR}SIR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [6.41]	0.17 [6.99]	0.13 [6.01]	0.39 [1.14]	0.14 [6.58]	0.11 [4.41]	0.16 [0.56]
SIR	0.31 [3.47]	0.39 [3.16]	0.34 [3.52]	0.50 [1.09]	0.39 [2.04]	0.50 [3.91]	-0.28 [-0.61]
Anomaly 1	0.27 [4.86]						0.86 [2.24]
Anomaly 2		0.41 [3.48]					0.14 [1.14]
Anomaly 3			0.13 [5.85]				-0.25 [-0.71]
Anomaly 4				0.11 [4.25]			-0.18 [-0.09]
Anomaly 5					0.10 [1.26]		-0.14 [-1.36]
Anomaly 6						0.71 [0.67]	0.73 [0.44]
# months	715	720	715	648	504	576	14
$\bar{R}^2(\%)$	0	0	0	2	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the SIR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{SIR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (5 year), Growth in book equity, Share issuance (1 year), Momentum and LT Reversal, Long-term EPS forecast, Analyst Value. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.19 [2.55]	0.27 [3.53]	0.26 [3.42]	0.25 [3.28]	0.27 [2.93]	0.25 [2.95]	0.20 [2.32]
Anomaly 1	32.35 [8.07]						12.10 [2.10]
Anomaly 2		25.53 [6.18]					22.50 [4.11]
Anomaly 3			26.21 [6.82]				15.58 [2.75]
Anomaly 4				4.49 [4.26]			3.49 [3.26]
Anomaly 5					-12.62 [-4.07]		-17.57 [-5.83]
Anomaly 6						-9.45 [-3.32]	-14.50 [-4.75]
mkt	3.74 [2.09]	3.28 [1.80]	2.95 [1.63]	2.45 [1.33]	-2.49 [-1.04]	5.49 [2.73]	-1.08 [-0.48]
smb	1.16 [0.44]	-1.97 [-0.75]	-0.01 [-0.00]	-2.82 [-1.05]	-1.47 [-0.43]	4.02 [1.30]	-2.83 [-0.88]
hml	-5.07 [-1.46]	-10.62 [-2.99]	-8.27 [-2.36]	-8.65 [-2.42]	-3.84 [-0.88]	-6.46 [-1.54]	6.89 [1.59]
rmw	-19.94 [-5.28]	-6.81 [-1.92]	-22.52 [-5.49]	-6.01 [-1.66]	2.37 [0.56]	5.70 [1.40]	-4.40 [-0.93]
cma	1.48 [0.27]	-8.09 [-1.24]	0.82 [0.14]	14.88 [2.83]	12.91 [1.98]	5.46 [0.95]	-22.10 [-3.03]
umd	0.13 [0.07]	1.80 [1.00]	1.06 [0.59]	-2.11 [-1.01]	-0.44 [-0.21]	0.91 [0.46]	-6.03 [-2.73]
# months	728	732	728	728	504	576	500
$\bar{R}^2(\%)$	10	7	8	4	5	5	21

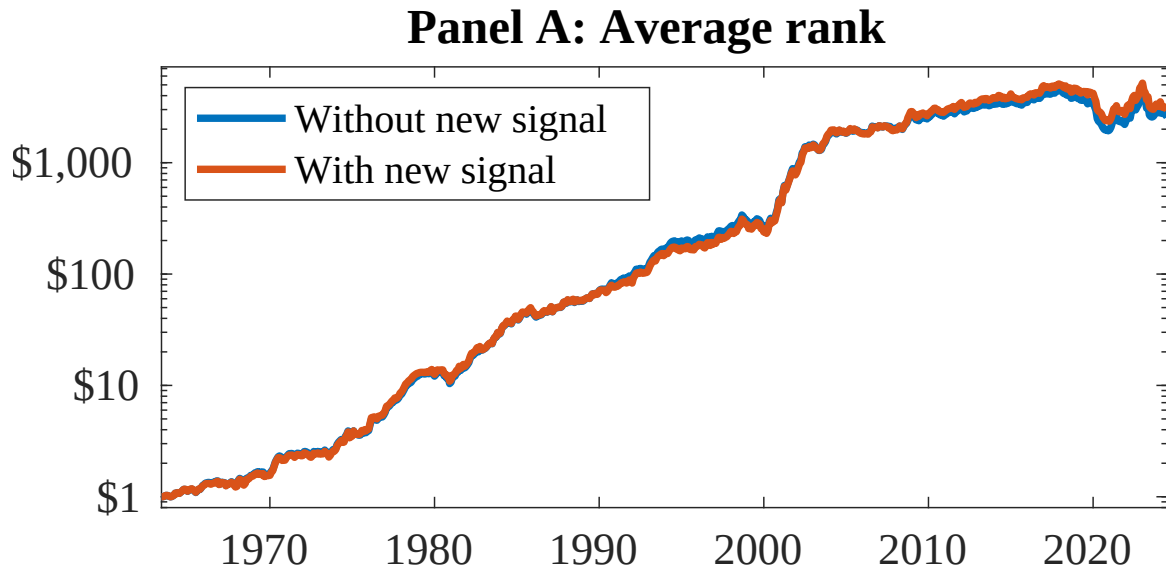


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as SIR. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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